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隱形帳簿：量化印尼平台經濟中的隱形楔子
The Invisible Ledger: Quantifying the Invisible Wedge in Indonesia's Platform Economy

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Abstract

This study measures the difference between the transaction value digital platforms process in Indonesia and the revenue those platforms recognize for themselves. I call that difference the invisible wedge and measure it with an Ecosystem Ratio, defined as unbooked gross transaction value divided by booked platform revenue. Because no Indonesian platform discloses a strictly country-labelled matched pair, every observation is retained under a named evidence tier and tiers are never pooled. Seventeen candidate platform-year levels across five issuer series spanning FY2019 to FY2025 show that the wedge moves, and disclosed revenue components explain the largest movements. Forty-eight matched issuer-years outside Indonesia corroborate that the boundary is general to platform intermediation while its magnitude depends on business model. Official statistics locate the wider activity beyond issuer accounts, and Indonesia's marketplace withholding rules show how the underlying records can enter third-party reporting.

1. Introduction

Southeast Asia's digital economy grew from \$100 billion in gross merchandise value in 2020 to \$263 billion in 2024 (Google, Temasek and Bain 2020, 2024). Much of that activity is coordinated through digital platforms, but the value of transactions a platform processes can be far larger than the revenue it recognizes for itself. These platforms record transactions, determine participant payouts, and book their own revenue. I call the difference between the transaction value processed through a platform and the revenue that platform books the “invisible wedge.”

Suppose a platform processes 100 units of transaction value but recognizes only 10 units as revenue. The remaining 90 units are still recorded within the platform's system but do not constitute platform revenue. Instead, they may represent merchant receipts, driver payouts, inventory costs, taxes paid elsewhere, and other pass-through payments. The paper measures this accounting gap without assuming that the residual represents profit, taxable income, or unpaid tax.

I measure the wedge with an ecosystem ratio defined as the gross transaction value (GTV) not recognized as platform revenue divided by the revenue that the platform does recognize. Inputs for Grab and Shopee are either Indonesia-specific or transparently derived for Indonesia. By contrast, Blibli and Bukalapak report both inputs directly but at a business or group scope wider than Indonesia alone. For fiscal year 2023 the three leading platform cases place transaction value not booked as platform revenue at approximately US\$40.07 billion, equivalent to roughly 2.9 percent of Indonesia's nominal gross domestic product. The paper measures gross participant-facing transaction value that is not booked as platform revenue; it does not estimate participant profit, unpaid tax, or value-added.

The term invisible refers to the administration's information channel, not to criminal concealment. Digital platform scale is not the same as tax-administrative visibility: the gross receipts earned by merchants, drivers, and other participants may remain outside routine third-party income reporting, even though the platform holds a record of the transaction that produced them. The wedge does not measure that reporting gap, and this paper does not claim it does. It measures the accounting boundary; whether the activity behind that boundary is separately recorded and transmitted to public institutions is a further question the study then examines.

1.1 Research questions

One question drives the proposal: how large is the invisible wedge in Indonesia, how does it change over time, and what explains those changes? Table 1 sets out three objectives answering it. A separate module of forty-eight matched issuer-years outside Indonesia provides external corroboration, testing whether the transaction–revenue boundary also appears across other platform business models; it is reported in Section 5 rather than as a research question of its own.

Table 1. Research objectives and principal evidence. RQ1 and RQ2 concern the measurement and its movement within Indonesia; RQ3 follows the activity and the underlying records beyond platform accounts. Global issuer evidence is reported as external corroboration in Section 5.

	Research question	Principal evidence
RQ1	How large is the invisible wedge in Indonesia, and does it persist across platforms and years?	17 retained candidate platform-year levels, three evidence tiers, FY2019–FY2025
RQ2	What explains movements in the wedge?	Disclosed revenue components: incentives, gross and net revenue, monetization, scope
RQ3	Where does activity outside platform revenue appear, and can the records reach administration?	BPS national, channel and recordkeeping evidence; PMK 37/2025 and DJP materials

1.2 Contributions

First, the study measures the wedge longitudinally. Earlier evidence for this market rests on a single fiscal year; the inventory assembled here spans seventeen retained candidate platform-years from FY2019 to FY2025, so the wedge's direction of travel can be reported and not only its scale.

Second, it explains why the wedge moves, reconciling the largest divergences against the revenue components issuers themselves disclose — monetization, customer incentives, gross-versus-net recognition, and reporting perimeter.

Third, it establishes whether the boundary is peculiar to Indonesia. Forty-eight matched issuer-years across eight non-Indonesian platform businesses provide external corroboration and turn the wide cross-platform spread in measured ratios from an obstacle into a result.

Fourth, it connects the accounting boundary to participant records and tax administration, using BPS evidence and Indonesia's PMK 37/2025 marketplace withholding architecture.

2. Literature review and research gap

2.1 Platform economics and the location of value

The economics of multisided platforms begins with the interaction of distinct user groups whose participation affects the value received by the other side. Rochet and Tirole (2003), Caillaud and Jullien (2003), Parker and Van Alstyne (2005), and Armstrong (2006) show why platform pricing cannot be read as a simple markup over cost: prices and commissions are chosen to balance cross-group network effects, participation, and competition. Hagiu and Wright (2015) further distinguish a multisided platform from a vertically integrated firm by locating decision rights and entrepreneurial effort with affiliated participants rather than inside the platform. These results explain why low retained shares can coexist with large ecosystems and why platform revenue is a poor proxy for the transaction value coordinated through the platform.

2.2 Revenue recognition and the accounting boundary

Accounting determines how much facilitated commerce becomes revenue. Under IFRS 15, principal-agent treatment governs gross-versus-net presentation, while customer incentives, advertising and service income, acquisitions, and reporting perimeter can move revenue without an equivalent change in commerce. De Franco, Kothari and Verdi (2011) show why comparability matters to users of financial statements.

2.3 Informality, platform work, and measurement

Traditional informal-economy estimates rely on household surveys, firm surveys, or indirect macroeconomic indicators. Medina and Schneider (2019) estimate a large global shadow economy; La Porta and Shleifer (2014) emphasise the small scale and low productivity of informal firms; and Ulysea (2018) shows that registration and off-the-books employment are distinct margins whose welfare effects cannot be inferred from formal status alone. Platform participants occupy an unusual position in this literature. Their transactions are priced, recorded, and settled through a formal intermediary, but the participant may remain self-employed, unregistered, below filing thresholds, or simply outside an automatic reporting channel. The object measured here is therefore not the shadow economy as conventionally defined. It is a digitally recorded flow whose administrative treatment resembles self-reported income even though an intermediary already possesses the record.

2.4 Third-party reporting as tax capacity

A central result in public finance is that tax compliance depends strongly on third-party information and remittance structure. Kleven et al. (2011) find an evasion rate of 0.3 percent on income subject to third-party reporting versus 37 percent on self-reported income in Denmark. Pomeranz (2015) shows that the VAT paper trail creates self-enforcement across firms, while Naritomi (2019) demonstrates that consumer-held records can raise reported sales. Kleven, Kreiner and Saez (2016) formalise firms as fiscal intermediaries, and Slemrod (2019) places information reporting and remittance regimes among the core instruments of modern enforcement. Platform work creates a new configuration — a large intermediary with detailed records but, in many jurisdictions, no comprehensive obligation to report seller or worker income. The administrative problem is therefore not a lack of data generation; it is the absence, incompleteness, or fragmentation of the reporting rule. Barrios, Hochberg, and Yi (2022) and Denes, Lagaras, and Tsoutsoura (2025) show that platform-mediated work leaves economically meaningful financial and administrative records.

2.5 Digital taxation and platform-reporting rules

Corporate digital-tax reforms and platform-participant reporting address different bases. The OECD Pillar One blueprint concerns the allocation of taxing rights over large multinational enterprises. By contrast, the OECD's Model Rules require platforms to collect and report information on income realised by sellers and service providers, and the European Union's DAC7 regime similarly imposes due-diligence and reporting duties on platform operators. These instruments provide the closest policy precedent for the mechanism studied here: using the platform's existing transaction record to improve participant-level visibility without treating platform revenue as the participant tax base. Indonesia's PMK 37/2025 applies the same logic through seller identification, transaction-linked reporting, and marketplace withholding.

2.6 Research gap

These literatures explain why transaction value exceeds platform revenue, why accounting moves the two apart, why participant records vary, and why third-party information matters. What is missing is a longitudinal, source-auditable account for one market that measures the wedge, explains why it moves, establishes whether the same boundary appears in platform businesses elsewhere, locates the wider digital activity beyond issuer accounts, and only then asks how the existing transaction records enter institutional reporting.

3. Theoretical framework: the ecosystem ratio

To capture the invisible wedge in Indonesia, I propose and construct a proxy, the ecosystem ratio. The measure is my own construction for this paper rather than one adopted from a prior study. For example, if a platform processes 100 units of transaction value and books 10 units as revenue, 90 units pass through the ecosystem without being recorded as the platform's own revenue. The ecosystem ratio formalizes that comparison: $(100 - 10) / 10 = 9$.

$$W = V - R \quad E = (V - R) / R \quad (1)$$

Gross transaction value (GTV), or gross merchandise value (GMV) in some platform disclosures, is the total value of transactions processed through a platform, regardless of who ultimately receives the proceeds. This construction is consistent with the platform-economics literature showing that booked platform revenue need not track the total transaction value coordinated through a multisided platform (Rochet and Tirole 2003; Evans and Schmalensee 2016). The ecosystem ratio describes the scale of participant-facing transaction value relative to the platform's auditable revenue base. A higher ratio indicates a larger unbooked transaction flow relative to the platform's own revenue; it does not by itself imply greater participant profit, taxable income, value-added, non-compliance, or tax due.

This distinction is essential. Platform filings and investor disclosures can make the accounting gap measurable, but the tax treatment of the residual varies according to participant type, deductible expenses, tax thresholds, registration status, and taxes already remitted through other channels. The empirical contribution of my paper is to quantify the scale of the recorded transaction flow that platforms process but do not book as their own revenue, and to establish how that quantity moves and why.

$$D = g(V) - g(R) \quad (2)$$

Because the transaction–revenue relationship also changes through time, the analysis compares transaction-value growth with platform-revenue growth for each within-series annual transition. Opposite signs identify periods in which activity and revenue move in different directions, and material movements are then reconciled against disclosed revenue components.

4. Data, methodology, and provenance

4.1 Sample construction and evidence tiers

Platform revenue and transaction value are taken from primary company disclosures. A platform-year is admitted only when both measures cover the same period and a scope that can be identified, units and definitions are known, derived inputs trace to source, and structural breaks are flagged rather than treated as ordinary year-to-year change. Repeated publication vintages of the same period are not counted as independent observations. All monetary values are reported in U.S. dollars. Take rate is defined as platform revenue divided by gross transaction value (GTV).

No Indonesian platform publishes a strictly country-labelled matched pair. Grab discloses Indonesia revenue but not country transaction value; Sea discloses no Indonesian figure; GoTo reports both only at Group level. Every observation is therefore admitted under an explicit tier and reported under that label, as Table 2 shows. Tokopedia FY2021 and Bukalapak FY2024 are excluded for period mismatch.

Table 2. Evidence tiers for Indonesian platform observations. Tokopedia is a business segment rather than a country line; Blibli 3P Retail also contains travel and Bukalapak reports at Group level with overseas operations; Grab and Shopee each require one side to be derived. Tiers are never pooled into a single monetary total.

Evidence tier	Series	Levels	Transitions
Direct Indonesia-aligned segment	Tokopedia e-commerce	2	1
Direct issuer, scope-pending	Blibli 3P Retail; Bukalapak Group	9	7
Conditional country reconstruction	Grab Indonesia; Shopee Indonesia	6	4
All tiers (inventory diagnostic)	5 series	17	12

4.2 Retained evidence modules

The study is not a single-fiscal-year design. Table 3 lists the seven retained modules, each with its own unit of observation. They are never summed into one sample size, because their geographic, regulatory and business-model definitions differ.

Table 3. Retained evidence modules and their role. Counts are not additive across rows.

Module	Coverage	Retained scale	Role
Direct Indonesian issuer candidates	FY2019–FY2025	13 period-matched; 12 with positive denominators	Candidate longitudinal core
Conditional country reconstructions	FY2021–FY2024	6 Grab/Shopee platform-years	Sensitivity only
Indonesia marketplace structure	2022–2025	Market-share and platform-structure evidence	Coverage and structural breaks
BPS official e-commerce evidence	2020–2024	National indicators; 39 provinces, 2023 and 2024	Activity, channels, recordkeeping
ASEAN corroboration	2019–2025	42 country-years from 450 source-vintage rows	Separate-country robustness
Global issuer corroboration	Multi-year	48 matched issuer-years, 8 businesses	Business-model corroboration
Historical quarterly accounting	2017–2022	47 company/segment-quarters	Historical disclosure evidence

4.3 Derivation of conditional country values

Grab's Indonesian transaction value is derived by applying its own Group monetization rate (Group revenue / Group GMV) to disclosed Indonesia revenue; Table 4 sets out the derivation year by year. The assumption is transparent and testable but is not a company disclosure. Shopee's Indonesian transaction value is an external market estimate from Momentum Works,

used because Sea discloses no Indonesian figure, labelled as third-party wherever it enters a calculation, and tested in Section 5.

Table 4. Worked derivation of conditional Indonesia transaction value for Grab. Indonesia revenue is directly disclosed and Group figures are source-reported; the final column is derived, not disclosed.

Year	Indonesia revenue	Group revenue	Group GMV	Conditional Indonesia GMV (derived)
FY2021	US\$79m	US\$675m	US\$16,061m	US\$1,879.7m
FY2022	US\$275m	US\$1,433m	US\$19,937m	US\$3,826.0m
FY2023	US\$605m	US\$2,359m	US\$20,983m	US\$5,381.4m

5. Preliminary evidence and feasibility

The assembled evidence already supports each step of the design. Table 5 reports the FY2023 cross-section: a combined wedge of US\$40.07 billion at an Ecosystem Ratio of 12.671. Varying each assumed parameter one at a time moves that wedge between US\$37.65 billion and US\$42.49 billion, and excluding each platform in turn leaves US\$35.29 billion, US\$24.14 billion and US\$20.70 billion respectively. No single assumption or platform drives the order of magnitude.

Table 5. Indonesia-focused FY2023 cross-section, US\$ billions. V is transaction value, R is platform revenue, E the Ecosystem Ratio. The final row sums three documented cases under mixed evidence classes and is not an Indonesia-wide total.

Case	V	R	Wedge	E	Evidence class
Grab Indonesia	5.381	0.605	4.776	7.895×	Derived V
Tokopedia e-commerce	16.331	0.405	15.926	39.296×	Direct pair
Shopee Indonesia	21.520	2.152	19.368	9.000×	Derived V and R
Selected platforms	43.233	3.162	40.070	12.671×	Sum, not a total

The wedge also moves. Table 6 reports two valid longitudinal summaries, differing only in which evidence is admitted. That the reversal count changes with admission is itself informative: it demonstrates on the study's own data that measurement choices can alter an empirical conclusion, which is why the sample hierarchy is shown rather than collapsed.

Table 6. Within-series annual growth divergence under two admission rules. Counts differ because evidence admission differs; both are reported rather than resolved into a single headline.

	All-tier inventory	Direct-candidate sensitivity
Series admitted	Blibli FY2021–25, Bukalapak, Tokopedia, Grab, Shopee	Blibli FY2020–25, Bukalapak, Tokopedia
Transitions	12	9
Revenue grows faster	10	6
Transaction grows faster	2	3
Sign reversals	2	3
Median absolute divergence	42.02 pp	42.94 pp

Mechanism is traceable. Tokopedia FY2022–FY2023 shows transaction value falling 8.90 percent while third-party net segment revenue rises 53.20 percent; reconciling the disclosed components, about 60.56 percent of the increase is associated with lower customer incentives and 39.44 percent with higher gross revenue, and the components sum exactly to reported net revenue in both years. Blibli 3P FY2022–FY2023 points the same way from a very low revenue base.

The pattern is not peculiar to Indonesia. Across 48 matched issuer-years for eight businesses — eBay, Etsy, Shopify, Jumia, Zalando, Rakuten, Mercado Libre and Sea — the boundary is present throughout while its level is business-model determined, with take rates from 0.22 to 74.63 percent and a median clean-scope growth divergence of 7.10 percentage points. Outside issuer accounts, BPS reports national e-commerce value rising 17.08 percent in 2024 while the marketplace component rises only 1.42 percent against 20.58 percent for non-marketplace, so roughly 98.49 percent of the nominal increase falls outside the channel platform accounts observe. PMK 37/2025 then supplies the institutional route: marketplaces designated 1 July 2026, collection effective 1 August, postponed through 31 October and now scheduled for 1 November 2026.

6. Limitations and advisor decisions

The wedge measures transaction value not booked as platform revenue. It does not establish that all participant-facing GTV is profit, taxable income, unpaid tax, or GDP value-added, and the paper does not estimate participant profit, unpaid tax, or value-added. No Indonesian platform publishes a strictly country-labelled matched pair, so every observation carries a named evidence tier and the tiers differ in what they support. Mechanism reconciliations are arithmetic against the issuer's own disclosed categories, not causal decompositions. BPS province evidence is ecological. PMK 37/2025 offers no completed implementation period from which to estimate a compliance effect.

Two decisions are sought at the proposal examination. First, whether the direct issuer scope-pending tier — Blibli 3P Retail and Bukalapak Group — enters the final longitudinal main sample, is reported only as a labelled tier, or is excluded. Second, whether the conditional country reconstructions for Grab and Shopee may appear in the main comparison or should be confined to sensitivity analysis. Both change the reported transition counts, as Table 6 shows, and neither can be settled by the data alone.

7. Work plan

Table 7. Planned work to the final defence. The empirical backend is assembled and source-linked; the remaining work is sample admission, requested analysis, and manuscript construction.

Period	Planned work
September 2026	Advisor review of the data package; proposal oral examination; freeze the sample-admission rule and the checks the committee requests.
October–November 2026	Complete requested reconciliations and source concordance; extend sensitivity to the approved tier boundary; freeze thesis tables and figures.
Before final defence	Rebuild the manuscript under the approved sample; retain supporting modules only where they advance the final argument.

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Primary and Institutional Sources

Data statement. Company filings, annual reports, prospectuses and results materials provide the platform data. BPS, Bank Indonesia, the Ministry of Finance and DJP publications provide the business, payment and policy evidence. The working data file records source, definition, publication date, currency and scope for every measure, so each figure can be checked against the original disclosure.

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Grab Holdings Limited; Sea Limited. Annual reports and Forms 20-F.

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Appendix A. Definition of Variables and Data Sources

Table A1. Variables used in the analysis, their definitions and sources. *V* and *R* are observed from issuer disclosures; *W*, *E*, the growth rates and *D* are derived from unrounded source inputs.

Variable	Definition	Source
<i>V</i>	Gross transaction value processed through the platform, at stated business and geographic scope	Issuer annual reports, prospectuses, results materials
<i>R</i>	Platform revenue recognized for the same period and scope	Issuer annual reports and financial statements
$W = V - R$	Absolute invisible wedge	Derived
$E = (V - R) / R$	Ecosystem Ratio: wedge relative to booked revenue	Derived
$g(V), g(R)$	Year-on-year growth in transaction value and revenue	Derived
$D = g(V) - g(R)$	Growth divergence	Derived
Take rate	Platform revenue divided by gross transaction value	Derived
Revenue components	Customer incentives, third-party gross and net revenue	Issuer segment notes where disclosed
Evidence tier	Direct Indonesia-aligned, direct scope-pending, or conditional	Admission rules, Section 4.1
E-commerce value / businesses	National nominal e-commerce value; estimated businesses	BPS-Statistics Indonesia